Data Format

Structured, Semi-structured and Unstructured Data

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BIG Data - 02

What is Data?

Tabular Data

columns = attributes for those observations

Rows = observations

Player	Minutes	Points	Rebounds	Assists
A	41	20	6	5
B	30	29	7	6
C	22	7	7	2
D	26	3	3	9
E	20	19	8	0
F	9	6	14	14
G	14	22	8	3
I	22	36	0	9
J	34	8	1	3

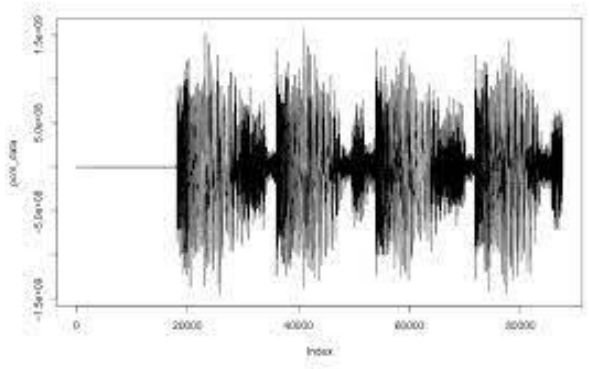
Log Data

```
weblogic.application.utils.StateMachineDriver.nextState(StateMachineDriver.java:26)
>
####<Dec 29, 2006 2:14:24 PM IST> <Notice> <Log Management> <svaidyan02> <xbusServer>
<[ACTIVE] ExecuteThread: '0' for queue: 'weblogic.kernel.Default (self-tuning)'> <<WLS
Kernel>> <> <> <1167381864275> <BEA-170027> <The server initialized the domain log
broadcaster successfully. Log messages will now be broadcasted to the domain log.>
####<Dec 29, 2006 2:14:24 PM IST> <Notice> <WebLogicServer> <svaidyan02> <xbusServer> <Main
Thread> <<WLS Kernel>> <> <> <1167381864976> <BEA-000365> <Server state changed to ADMIN>
####<Dec 29, 2006 2:14:24 PM IST> <Notice> <WebLogicServer> <svaidyan02> <xbusServer> <Main
Thread> <<WLS Kernel>> <> <> <1167381864996> <BEA-000365> <Server state changed to RESUMING>
####<Dec 29, 2006 2:14:28 PM IST> <Notice> <Security> <svaidyan02> <xbusServer> <[STANDBY]
ExecuteThread: '5' for queue: 'weblogic.kernel.Default (self-tuning)'> <<WLS Kernel>> <> <>
<1167381868541> <BEA-090171> <Loading the identity certificate and private key stored under
the alias DemoIdentity from the jks keystore file
C:\bea2613a\WEBLOG~1\server\lib\DemoIdentity.jks.>
####<Dec 29, 2006 2:14:29 PM IST> <Notice> <Security> <svaidyan02> <xbusServer> <[STANDBY]
ExecuteThread: '5' for queue: 'weblogic.kernel.Default (self-tuning)'> <<WLS Kernel>> <> <>
<1167381869643> <BEA-090169> <Loading trusted certificates from the jks keystore file
C:\bea2613a\WEBLOG~1\server\lib\DemoTrust.jks.>
####<Dec 29, 2006 2:14:29 PM IST> <Notice> <Security> <svaidyan02> <xbusServer> <[STANDBY]
ExecuteThread: '5' for queue: 'weblogic.kernel.Default (self-tuning)'> <<WLS Kernel>> <> <>
<1167381869713> <BEA-090169> <Loading trusted certificates from the jks keystore file
C:\bea2613a\JROCKI~1\jre\lib\security\cacerts.>
####<Dec 29, 2006 2:15:32 PM IST> <Warning> <Server> <svaidyan02> <xbusServer>
<DynamicSSLListenThread[DefaultSecure[1]]> <<WLS Kernel>> <> <> <1167381932743> <BEA-002611>
<Hostname "svaidyan02.apac.bea.com", maps to multiple IP addresses: 192.168.1.5,
172.22.56.120>
####<Dec 29, 2006 2:15:32 PM IST> <Notice> <Server> <svaidyan02> <xbusServer> <[STANDBY]
ExecuteThread: '5' for queue: 'weblogic.kernel.Default (self-tuning)'> <<WLS Kernel>> <> <>
<1167381932753> <BEA-002613> <Channel "Default[2]" is now listening on 127.0.0.1:7021 for
```



Forex Data

Audio Data



Questionnaire

```
RespondentId,StartDate,CompletedDate,LanguageCode,Question1,Question2,Question3,Question4,Question5,Question6,Question7,Questi
27357,2006.11.27 15:6,2006.11.27 15:7,en,Denmark,Financial Services,6 - 12 months,26-100,4,4,2,"cvbcvb",2,3,3,1,0pinio,1,0,0,1
27359,2006.11.27 15:7,2006.11.27 15:8,en,Italy,Hardware Vendor,1 - 2 years,26-100,3,5,4,,1,3,3,4,0pinio,0,0,0,0,1,0,0,1,0,,,,0
27360,2006.11.27 15:8,2006.11.27 15:8,en,Lithuania,Retail,6 - 12 months,6-10,4,1,4,"this is a random other text",2,2,2,2,0pini
27361,2006.11.27 15:8,2006.11.27 15:8,en,Panama,Retail,6 - 12 months,6-10,4,1,4,"this is a random other text",2,2,2,2,0pinio,0
27362,2006.11.27 15:8,2006.11.27 15:8,en,Djibouti,Manufacturing,6+ years,101-250,0,4,0,"another random text",5,5,5,5,0pinio,1,
27363,2006.11.27 15:8,2006.11.27 15:8,en,Tanzania,Retail,1 - 2 years,1001-5000,1,1,1,"123456",2,2,2,2,0pinio,0,1,1,1,1,1,1,1,1
27364,2006.11.27 15:8,2006.11.27 15:8,en,Vanuatu,Other,1 - 2 years,1001-5000,6,5,6,"123456",6,6,6,6,0pinio,0,0,1,1,1,1,0,1,1,"
27365,2006.11.27 15:8,2006.11.27 15:8,en,Angola,Government,1 - 2 years,11-25,4,2,4,"123456",3,3,3,3,0pinio,0,0,1,1,1,1,1,1,0,,
27366,2006.11.27 15:8,2006.11.27 15:8,en,Panama,Manufacturing,<6 months,1-5,1,4,1,"hey",5,5,5,5,0pinio,0,1,0,0,0,1,0,0,0,,"hey
27367,2006.11.27 15:8,2006.11.27 15:8,en,Norway,Education,2 - 5 years,5001-10000,6,0,6,"£6{[]}+&æ¤´´´*-/+\",1,1,1,1,0pinio,1,
27368,2006.11.27 15:8,2006.11.27 15:8,en,Bermuda,Software Vendor,1 - 2 years,11-25,0,2,0,"123456",3,3,3,3,0pinio,1,0,1,0,0,1,0
27369,2006.11.27 15:8,2006.11.27 15:8,en,Panama,Transportation,1 - 2 years,11-25,5,4,5,"123456",5,5,5,5,0pinio,0,1,0,0,0,1,0,0
27370,2006.11.27 15:8,2006.11.27 15:8,en,Maldives,Other,6+ years,10001 or more,2,5,2,"another random text",6,6,6,6,Network Pro
27371,2006.11.27 15:8,2006.11.27 15:8,en,Kyrgyzstan,Medical,2 - 5 years,26-100,3,5,3,"£6{[]}+&æ¤´´´*-/+\",6,6,6,6,Network Pro
27372,2006.11.27 15:8,2006.11.27 15:8,en,Antigua and Barbuda,Government,6 - 12 months,501-1000,6,2,6,"this is a random other t
27373,2006.11.27 15:8,2006.11.27 15:8,en,Belarus,Financial Services,6+ years,10001 or more,2,1,2,"another random text",2,2,2,2
27374,2006.11.27 15:8,2006.11.27 15:8,en,Vatican City,Non-profit,1 - 2 years,11-25,0,0,0,"123456",1,1,1,1,Network Probe,1,0,0,
27375,2006.11.27 15:8,2006.11.27 15:8,en,Georgia,Financial Services,6+ years,10001 or more,6,1,6,"another random text",2,2,2,2
27376,2006.11.27 15:8,2006.11.27 15:8,en,Tokelau,Transportation,1 - 2 years,11-25,2,4,2,"123456",5,5,5,5,Network Probe,0,1,0,0
27377,2006.11.27 15:8,2006.11.27 15:8,en,Chad,Software Vendor,<6 months,1-5,6,2,6,"hey",3,3,3,3,Network Probe,1,1,1,1,1,1,1,1,
27378,2006.11.27 15:8,2006.11.27 15:8,en,Turkey,Software Vendor,6 - 12 months,501-1000,1,2,1,"this is a random other text",3,3
27379,2006.11.27 15:8,2006.11.27 15:8,en,East Timor,Transportation,<6 months,1-5,0,4,0,"hey",5,5,5,5,0pinio,1,1,0,0,1,0,1,1,0,
27380,2006.11.27 15:8,2006.11.27 15:8,en,Nicaragua,Medical,6 - 12 months,6-10,5,5,5,"this is a random other text",6,6,6,6,0pin
27381,2006.11.27 15:8,2006.11.27 15:8,en,Equatorial Guinea,Software Vendor,6+ years,101-250,6,2,6,"another random text",3,3,3,
27382,2006.11.27 15:8,2006.11.27 15:8,en,Zambia,Retail,<6 months,251-500,1,1,1,"hey",2,2,2,2,Surveyor,0,1,0,0,0,0,0,1,0,,"hey"
27383,2006.11.27 15:8,2006.11.27 15:8,en,French Southern and Antarctic Lands,Retail,1 - 2 years,1001-5000,2,1,2,"123456",2,2,2
27384,2006.11.27 15:8,2006.11.27 15:8,en,Guinea-Bissau,Hardware Vendor,2 - 5 years,26-100,6,3,6,"£6{[]}+&æ¤´´´*-/+\",4,4,4,4,
27385,2006.11.27 15:8,2006.11.27 15:8,en,Viet Nam,Medical,2 - 5 years,26-100,4,5,4,"£6{[]}+&æ¤´´´*-/+\",6,6,6,6,0pinio,1,1,1,
27386,2006.11.27 15:8,2006.11.27 15:8,en,Reunion,Medical,1 - 2 years,1001-5000,2,5,2,"123456",6,6,6,6,0pinio,1,1,1,1,1,1,1,1,1
27387,2006.11.27 15:8,2006.11.27 15:8,en,Puerto Rico,Non-profit,<6 months,1-5,0,0,0,"hey",1,1,1,1,0pinio,1,1,1,1,0,1,1,1,0,,"h
27388,2006.11.27 15:8,2006.11.27 15:8,en,East Timor,Financial Services,6 - 12 months,6-10,1,1,1,"this is a random other text",
27389,2006.11.27 15:8,2006.11.27 15:8,en,Northern Mariana Islands,Software Vendor,<6 months,1-5,2,2,2,"hey",3,3,3,3,0pinio,1,0
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Data Types

Structured data –

Structured data is data whose elements are addressable for effective analysis. They have relational keys and can easily be mapped into pre-designed fields.

Semi-Structured data –

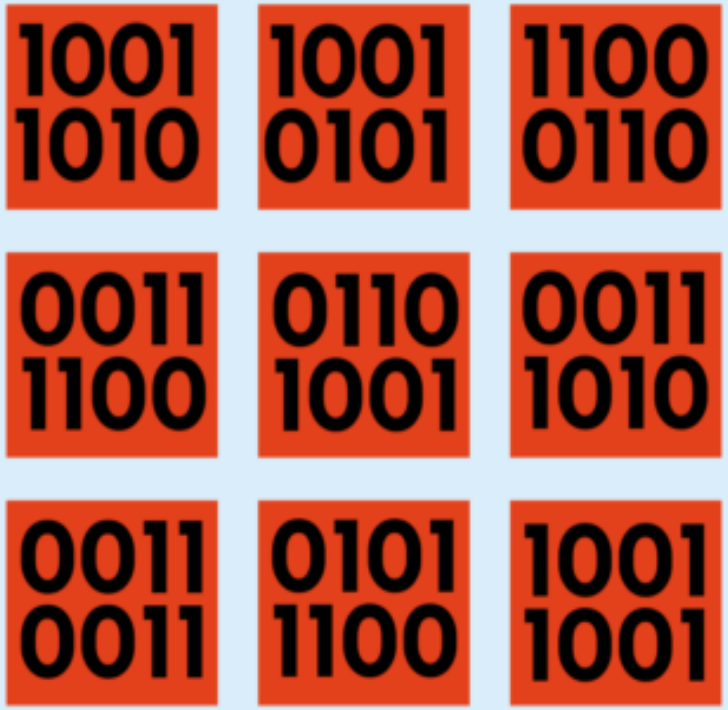
Semi-structured data is information that does not reside in a relational database but that has some organizational properties that make it easier to analyze.

Unstructured data –

Unstructured data is a data which is not organized in a predefined manner or does not have a predefined data model, thus it is not a good fit for a mainstream relational database.

Types of Big Data

Structured



Unstructured



Semi-Structured



Properties	Structured data	Semi-structured data	Unstructured data
Technology	It is based on Relational database table	It is based on XML/RDF(Resource Description Framework).	It is based on character and binary data
Transaction management	Matured transaction and various concurrency techniques	Transaction is adapted from DBMS not matured	No transaction management and no concurrency
Version management	Versioning over tuples,row,tables	Versioning over tuples or graph is possible	Versioned as a whole
Flexibility	It is schema dependent and less flexible	It is more flexible than structured data but less flexible than unstructured data	It is more flexible and there is absence of schema
Scalability	It is very difficult to scale DB schema	It's scaling is simpler than structured data	It is more scalable.
Robustness	Very robust	New technology, not very spread	–
Query performance	Structured query allow complex joining	Queries over anonymous nodes are possible	Only textual queries are possible

Attribute – Features Data

What is Data?

- Collection of data objects and their attributes
- An attribute is a property or characteristic of an object
  - Examples: eye color of a person, temperature, etc.
  - Attribute is also known as variable, field, characteristic, or feature
- A collection of attributes describe an object
  - Object is also known as record, point, case, sample, entity, or instance

Attributes

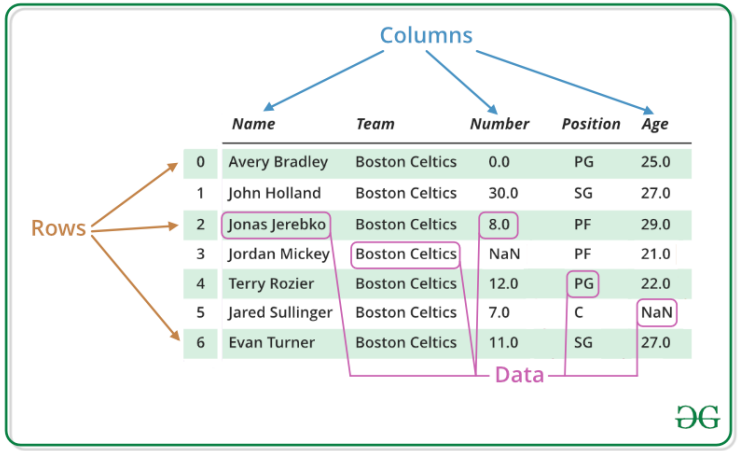
Objects

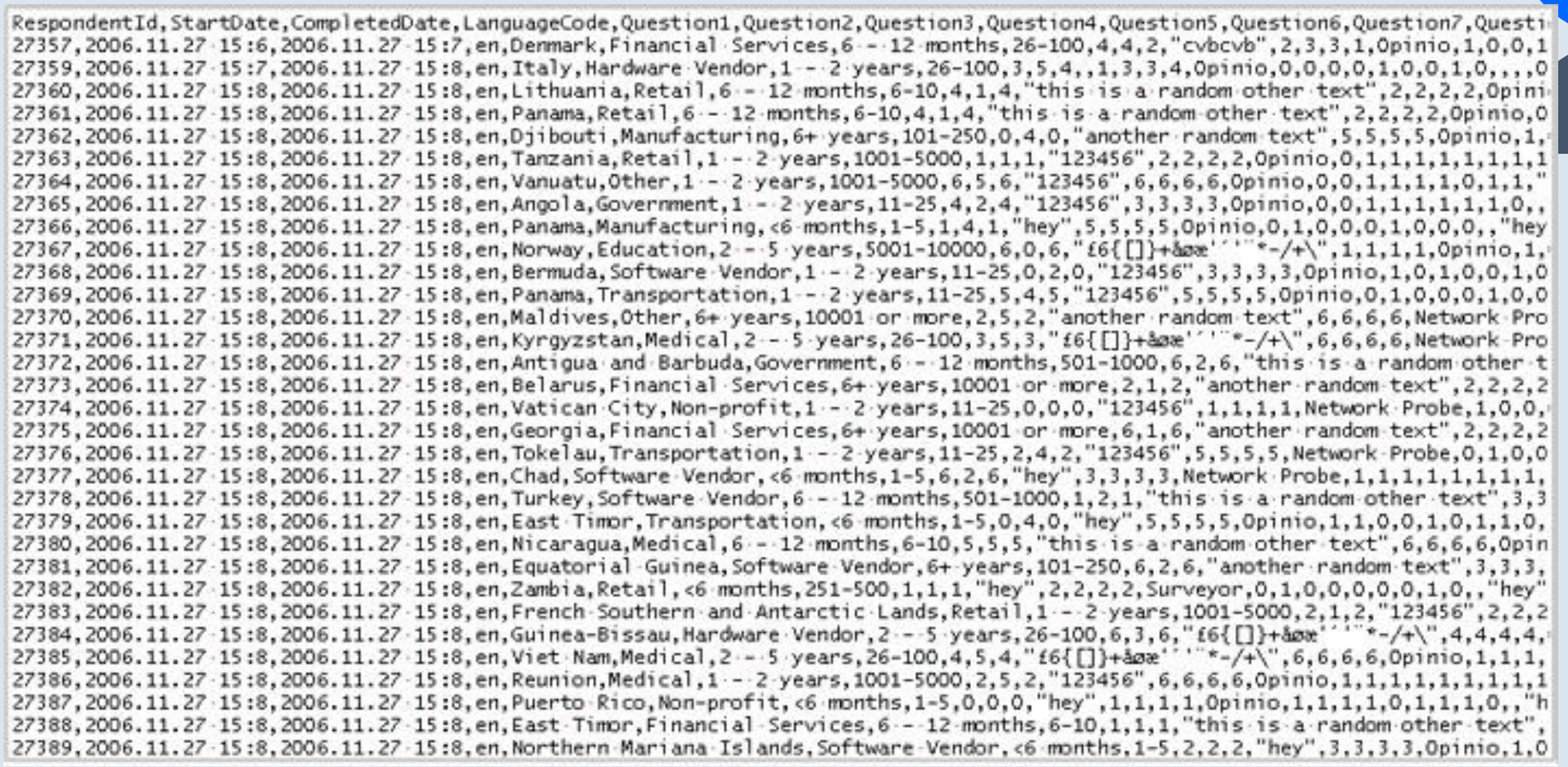
Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Attributes

- **Attribute (**or **dimensions, features, variables)**: a data field, representing a characteristic or feature of a data object.
  - *E.g., customer _ID, name, address*
- Types:
  - Nominal
  - Binary
  - Numeric: quantitative
    - Interval-scaled
    - Ratio-scaled

6





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RespondentId,StartDate,CompletedDate,LanguageCode,Question1,Question2,Question3,Question4,Question5,Question6,Question7,Questi
27357,2006.11.27 15:6,2006.11.27 15:7,en,Denmark,Financial Services,6 - 12 months,26-100,4,4,2,"cvbcvb",2,3,3,1,Opinio,1,0,0,1
27359,2006.11.27 15:7,2006.11.27 15:8,en,Italy,Hardware Vendor,1 - 2 years,26-100,3,5,4,,1,3,3,4,Opinio,0,0,0,0,1,0,0,1,0,,,,0
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27363,2006.11.27 15:8,2006.11.27 15:8,en,Tanzania,Retail,1 - 2 years,1001-5000,1,1,1,"123456",2,2,2,2,Opinio,0,1,1,1,1,1,1,1,1
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27368,2006.11.27 15:8,2006.11.27 15:8,en,Bermuda,Software Vendor,1 - 2 years,11-25,0,2,0,"123456",3,3,3,3,Opinio,1,0,1,0,0,1,0
27369,2006.11.27 15:8,2006.11.27 15:8,en,Panama,Transportation,1 - 2 years,11-25,5,4,5,"123456",5,5,5,5,Opinio,0,1,0,0,0,1,0,0
27370,2006.11.27 15:8,2006.11.27 15:8,en,Maldives,Other,6+ years,10001 or more,2,5,2,"another random text",6,6,6,6,Network Pro
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27374,2006.11.27 15:8,2006.11.27 15:8,en,Vatican City,Non-profit,1 - 2 years,11-25,0,0,0,"123456",1,1,1,1,Network Probe,1,0,0,
27375,2006.11.27 15:8,2006.11.27 15:8,en,Georgia,Financial Services,6+ years,10001 or more,6,1,6,"another random text",2,2,2,2
27376,2006.11.27 15:8,2006.11.27 15:8,en,Tokelau,Transportation,1 - 2 years,11-25,2,4,2,"123456",5,5,5,5,Network Probe,0,1,0,0
27377,2006.11.27 15:8,2006.11.27 15:8,en,Chad,Software Vendor,<6 months,1-5,6,2,6,"hey",3,3,3,3,Network Probe,1,1,1,1,1,1,1,1,
27378,2006.11.27 15:8,2006.11.27 15:8,en,Turkey,Software Vendor,6 - 12 months,501-1000,1,2,1,"this is a random other text",3,3
27379,2006.11.27 15:8,2006.11.27 15:8,en,East Timor,Transportation,<6 months,1-5,0,4,0,"hey",5,5,5,5,Opinio,1,1,0,0,1,0,1,1,0,
27380,2006.11.27 15:8,2006.11.27 15:8,en,Nicaragua,Medical,6 - 12 months,6-10,5,5,5,"this is a random other text",6,6,6,6,Opin
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27383,2006.11.27 15:8,2006.11.27 15:8,en,French Southern and Antarctic Lands,Retail,1 - 2 years,1001-5000,2,1,2,"123456",2,2,2
27384,2006.11.27 15:8,2006.11.27 15:8,en,Guinea-Bissau,Hardware Vendor,2 - 5 years,26-100,6,3,6,"£6{[]}+åøæ'´`"*-/+\",4,4,4,4,
27385,2006.11.27 15:8,2006.11.27 15:8,en,Viet Nam,Medical,2 - 5 years,26-100,4,5,4,"£6{[]}+åøæ'´`"*-/+\",6,6,6,6,Opinio,1,1,1,
27386,2006.11.27 15:8,2006.11.27 15:8,en,Reunion,Medical,1 - 2 years,1001-5000,2,5,2,"123456",6,6,6,6,Opinio,1,1,1,1,1,1,1,1,1
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27388,2006.11.27 15:8,2006.11.27 15:8,en,East Timor,Financial Services,6 - 12 months,6-10,1,1,1,"this is a random other text",
27389,2006.11.27 15:8,2006.11.27 15:8,en,Northern Mariana Islands,Software Vendor,<6 months,1-5,2,2,2,"hey",3,3,3,3,Opinio,1,0
```

Numerical, Categorical and Ordinal Data

Numerical numerical measurement expressed not by means of a natural language description, but rather in terms of numbers

Categorical measurement expressed not in terms of numbers, but rather by means of a natural language description

Ordinal mixes numerical and categorical data.

Data- Data can be classified into two different types.

Categorical	**Numerical**
Values or observations that can be sorted into groups or categories. Bar charts and pie graphs are used to graph categorical data.	Values or observations that can be measured. And these numbers can be placed in ascending or descending order. Scatter plots and line graphs are used to graph numerical data.

Nominal	**Ordinal**	**Discrete**	**Continuous**
Values or observations can be assigned a code in the form of a number where the numbers are simply labels. You can **count but not order** or measure nominal data. Examples: Sex, and eye colour.	Values or observations can be ranked (put in order) or have a rating scale attached. You can **count and order**, but not measure, ordinal data. Example: house numbers and swimming level.	Values or observations that is counted as **distinct and separate** and can only take particular values. Examples: the number of kittens in a litter; number of threads in a sheet, number of stars given for an energy rating.	You can **measure** continuous data. Values or observations **may take on any value** within a finite or infinite interval. Examples: height, time and temperature.

Attribute or Feature Types

Attribute Level	Transformation	Comments
Nominal	Any permutation of values	If all employee ID numbers were reassigned, would it make any difference?
Ordinal	An order preserving change of values, i.e., *new_value = f(old_value)* where *f* is a monotonic function.	An attribute encompassing the notion of good, better best can be represented equally well by the values {1, 2, 3} or by { 0.5, 1, 10}.
Interval	*new_value =a * old_value + b* where a and b are constants	Thus, the Fahrenheit and Celsius temperature scales differ in terms of where their zero value is and the size of a unit (degree).
Ratio	*new_value = a * old_value*	Length can be measured in meters or feet.

Scale	Basic Comparison	Measures of typical examples	Average
Nominal	Identity	Male/Female	Mode
		User/Non user	
		Occupations	
		Uniform numbers	
Ordinal	Order	Brand preference	Median
		Social class	
		Hardness of minerals	
		Graded quality of lumber	
Interval	Comparison of intervals	Temperature scale	Mean
		Grade point average	
		Attitude towards brand	
Ratio	Comparison of absolute magnitudes	Units sold	Geometric mean & Harmonic mean
		Number of purchases	
		Income	
		Age	

Feature Scaling

- In practice, different types of variables are often encountered in the same data set.
- A significant issue is that the range of the variables may differ a lot
  - Standarization

    The result of standardization (or Z-score normalization) is that the features will be rescaled to ensure the mean and the standard deviation are 0 and 1, respectively
  - Normalization

    This technique is to re-cales features with a distribution value between 0 and 1

Missing Value

sample_flow

		StartTime	Dur	Proto	SrcAddr	Sport
2	0	2011/08/10 09:46:59.607825	1.026539	tcp	94.44.127.113	1577
3	1	2011/08/10 09:47:00.634364	1.009595	tcp	94.44.127.113	1577
4	2	2011/08/10 09:47:48.185538	3.056586	tcp	147.32.86.89	4768
5	3	2011/08/10 09:47:48.230897	3.111769	tcp	147.32.86.89	4788
6	4	2011/08/10 09:47:48.963351	3.083411	tcp	147.32.86.89	4850
7	5	2011/08/10 09:47:58.806814	3.097288	tcp	147.32.86.89	4866
8	6	2011/08/10 09:51:34.450457	1.048908	tcp	213.200.244.217	47908
9	7	2011/08/10 09:54:55.231320	4.373526	tcp	75.105.28.60	1419
10	8	2011/08/10 09:57:13.352114	4.827912	tcp	75.105.28.60	1491
11	9	2011/08/10 09:58:43.301515	0.049697	tcp	178.111.79.115	41752
12	10	2011/08/10 09:54:09.710772	328.361664	tcp	147.32.84.59	49185
13	11	2011/08/10 10:00:34.864769	5.242459	tcp	75.105.28.60	1586
14	12	2011/08/10 10:01:16.344485	0.972390	tcp	89.31.40.106	28451
15	13	2011/08/10 10:06:19.661695	0.923098	tcp	89.31.40.106	13717
16	14	2011/08/10 10:07:41.514293	1.009763	tcp	188.112.70.72	1817
17	15	2011/08/10 10:08:18.464075	0.969967	tcp	85.248.56.40	42480
18	16	2011/08/10 10:09:36.758829	2.853907	tcp	41.188.145.202	2285
19	17	2011/08/10 10:09:38.376337	2.948305	tcp	41.188.145.202	2288
20	18	2011/08/10 10:09:39.984069	2.956415	tcp	41.188.145.202	2291
21	19	2011/08/10 10:09:44.359242	0.920374	tcp	89.31.40.106	39927
22	20	2011/08/10 10:09:39.612736	6.031078	tcp	41.188.145.202	2285
23	21	2011/08/10 10:09:41.324642	6.035653	tcp	41.188.145.202	2288
24	22	2011/08/10 10:09:42.940484	6.028175	tcp	41.188.145.202	2291
25	23	2011/08/10 10:09:51.394331	0.000000	tcp	147.32.84.59	6881
26	24	2011/08/10 10:09:57.619871	1.383731	tcp	213.24.237.172	15007
27	25	2011/08/10 10:10:53.596635	0.939583	tcp	85.248.56.40	42572
28	26	2011/08/10 10:13:28.886935	3.015055	tcp	69.231.198.54	4144
29	27	2011/08/10 10:13:31.657243	0.987838	tcp	188.112.70.72	2057
30	28	2011/08/10 10:13:42.930890	1.419550	tcp	95.153.177.13	35049
31	29	2011/08/10 10:13:47.285295	3.024265	tcp	69.231.198.54	4154
32	..	...	...	...	...	...

sample_flow | ctu12_capture20110819.binetflow

```
StartTime,Dur,Proto,SrcAddr,Sport,Dir,DstAddr,Dport,State,sTos,dTos,TotPkts,TotBytes,SrcBytes,Label
2011/08/19 10:32:31.926442,3.003373,tcp,147.32.86.194,4207,   ->,77.75.73.9,80,SR_A,0,0,3,182,122,flow=Background-TCP-Attempt
2011/08/19 10:37:05.849782,0.967492,tcp,147.32.3.51,1056,   ->,147.32.87.22,10010,S_RA,0,0,4,244,124,flow=Background-TCP-Attempt
2011/08/19 10:38:48.257124,0.000000,tcp,60.2.174.102,80,   ?>,147.32.85.48,21885,RA_,0,,1,60,60,flow=Background
2011/08/19 10:41:12.342795,1.391385,tcp,200.99.228.67,3259,   ->,147.32.87.62,10401,S_RA,0,0,4,244,124,flow=Background-TCP-Attempt
2011/08/19 10:44:52.276736,1.124664,tcp,147.32.3.51,1089,   ->,147.32.84.46,10010,S_RA,0,0,4,244,124,flow=Background-TCP-Attempt
2011/08/19 10:46:23.837244,0.999620,tcp,147.32.3.51,1095,   ->,147.32.87.22,10010,S_RA,0,0,4,244,124,flow=Background-TCP-Attempt
2011/08/19 10:48:35.684493,1.712786,tcp,116.12.223.151,64689,   ->,147.32.84.201,443,S_RA,0,0,4,252,132,flow=Background-TCP-Attempt
2011/08/19 10:49:00.271166,1.093437,tcp,147.32.3.51,1106,   ->,147.32.84.46,10010,S_RA,0,0,4,244,124,flow=Background-TCP-Attempt
2011/08/19 10:49:29.833210,1.155755,tcp,147.32.3.51,1108,   ->,147.32.87.22,10010,S_RA,0,0,4,244,124,flow=Background-TCP-Attempt
2011/08/19 10:50:31.831637,0.952887,tcp,147.32.3.51,1112,   ->,147.32.87.22,10010,S_RA,0,0,4,244,124,flow=Background-TCP-Attempt
2011/08/19 10:51:33.830199,1.077859,tcp,147.32.3.51,1118,   ->,147.32.87.22,10010,S_RA,0,0,4,244,124,flow=Background-TCP-Attempt
2011/08/19 10:55:12.262794,1.280871,tcp,147.32.3.51,1132,   ->,147.32.84.46,10010,S_RA,0,0,4,244,124,flow=Background-TCP-Attempt
2011/08/19 10:57:45.821794,1.077884,tcp,147.32.3.51,1142,   ->,147.32.87.22,10010,S_RA,0,0,4,244,124,flow=Background-TCP-Attempt
2011/08/19 10:58:06.872596,2.671495,tcp,195.94.242.147,49539,   ->,147.32.84.202,443,SR_SA,0,0,3,184,122,flow=Background-TCP-Established
2011/08/19 10:58:09.544091,5.470439,tcp,195.94.242.147,49539,   ->,147.32.84.202,443,SR_SA,0,0,3,184,122,flow=Background-TCP-Established
2011/08/19 10:58:13.772620,2.755254,tcp,195.94.242.147,49592,   ->,147.32.84.202,443,SR_SA,0,0,3,184,122,flow=Background-TCP-Established
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2011/08/19 10:58:50.367119,2.245816,tcp,217.100.175.20,4411,   ->,147.32.84.201,80,S_RA,0,0,4,276,156,flow=Background-TCP-Attempt
2011/08/19 10:58:52.612935,2.201548,tcp,217.100.175.20,4411,   ->,147.32.84.201,80,S_RA,0,0,4,276,156,flow=Background-TCP-Attempt
2011/08/19 10:59:58.287834,2.333980,tcp,217.100.175.20,4451,   ->,147.32.84.201,443,S_RA,0,0,4,276,156,flow=Background-TCP-Attempt
2011/08/19 11:00:00.621814,1.629306,tcp,217.100.175.20,4451,   ->,147.32.84.201,443,S_RA,0,0,4,276,156,flow=Background-TCP-Attempt
2011/08/19 10:38:36.775411,1322.733643,tcp,202.59.157.38,1225,   ->,147.32.87.44,80,FSRPA_SPA,0,0,29,17775,939,flow=Background-TCP-Established
2011/08/19 11:00:43.232849,2.871080,tcp,147.32.86.58,2444,   ->,77.75.73.9,80,SR_A,0,0,3,182,122,flow=Background-TCP-Attempt
2011/08/19 10:49:02.309056,708.479187,tcp,147.32.84.59,49193,   ->,147.32.80.7,80,FSRPA_FSPA,0,0,29,8426,1678,flow=Background-Established-cmpgw-CVUT
2011/08/19 11:00:41.242570,9.919378,tcp,50.129.93.15,53912,   ->,147.32.84.229,80,SR_SA,0,0,9,660,294,flow=Background-TCP-Established
2011/08/19 11:01:53.817329,1.156492,tcp,147.32.3.51,1162,   ->,147.32.87.22,10010,S_RA,0,0,4,244,124,flow=Background-TCP-Attempt
2011/08/19 11:05:10.880875,8.995331,tcp,84.203.152.100,48080,   ->,147.32.84.229,443,SR_SA,0,0,7,424,184,flow=Background-TCP-Established
2011/08/19 11:09:02.659025,1.198686,tcp,62.24.111.246,51788,   ->,147.32.84.229,13363,SR_SA,0,0,3,216,138,flow=Background-TCP-Established
2011/08/19 11:09:03.857711,1.100881,tcp,62.24.111.246,51788,   ->,147.32.84.229,13363,SR_SA,0,0,3,216,138,flow=Background-TCP-Established
2011/08/19 11:09:04.620430,1.074600,tcp,62.24.111.246,51799,   ->,147.32.84.229,443,SR_SA,0,0,3,216,138,flow=Background-TCP-Established
2011/08/19 11:09:06.324829,1.565212,tcp,62.24.111.246,51818,   ->,147.32.84.229,80,SR_SA,0,0,3,216,138,flow=Background-TCP-Established
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2011/08/19 11:09:08.900242,0.477685,tcp,62.24.111.246,51799,   ->,147.32.84.229,443,SR_SA,0,0,3,216,138,flow=Background-TCP-Established
2011/08/19 11:09:08.459364,1.359952,tcp,62.24.111.246,51788,   ->,147.32.84.229,13363,SR_SA,0,0,3,216,138,flow=Background-TCP-Established
2011/08/19 11:09:09.234543,0.973901,tcp,62.24.111.246,51818,   ->,147.32.84.229,80,SR_SA,0,0,3,216,138,flow=Background-TCP-Established
2011/08/19 11:10:18.797266,7.500037,tcp,196.201.134.177,49583,   ->,147.32.84.184,443,SRA_,0,,7,420,420,flow=Background-TCP-Attempt
2011/08/19 11:12:43.611756,3.058666,tcp,62.28.8.186,49554,   ->,147.32.84.229,13363,SR_SA,0,0,3,180,120,flow=Background-TCP-Established
2011/08/19 11:12:45.262236,2.938219,tcp,62.28.8.186,49558,   ->,147.32.84.229,443,SR_SA,0,0,3,180,120,flow=Background-TCP-Established
2011/08/19 11:12:46.670422,6.001781,tcp,62.28.8.186,49554,   ->,147.32.84.229,13363,SR_SA,0,0,3,180,120,flow=Background-TCP-Established
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sample_flow | ctu12_capture20110819.binetflow | creditcard.csv

```
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0,1.191857111314 86,0.266150712059 63,0.166480113353 21,0.448154078460911,0.0600176492822243,-0.0823608088155687,-0.0788029833323113,0.0851016549148104,-0.255425128109186,-0.166974414004614,1.61272666105479,1.065235311372 87,0.489095015896 08,-0.143772296441519,0.635558093258208,0.463917041022171,-0.114804663102346,-0.183361270123994,-0.145783041325259,-0.0690831352230203,-0.225775248033138,-0.638671952771851,0.101288021253234,-0.339846475529127,0.167170404418143,0.125894532368176,-0.00898309914322813,0.0147241691924927,2.69,"0"
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How to handling missing value

- Ignore the data row
- Use a global constant to fill in for missing values
- Use attribute mean
- Use attribute mean for all samples belonging to the same class
- Use a data mining algorithm to predict the most probable value

Thank you

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